

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: F. Y. B. Tech.****Semester :I** Branch: Common to all branches**Subject Code & Name: BTBS 102/ (Engineering Chemistry)****Max Marks: 60****Date: 08/02/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Questions)		12
1	Ca-EDTA is a ----- complex.	1	1
	a. blue colour b. colourless c. red colour d. pink colour		
2	The unit of Hardness is -----	1	1
	a. ppm b. Nm c. mhos d. ohm		
3	The water which produce lather with soap is called ----- water	1	1
	a. Hard b. Soft c. bleached d. salty		
4	Sulphur system is a ----- component system.	2	1
	a. three b. two c. One d. zero		
5	Along vapour pressure curve of water system.....phases are in equilibrium.	2	1
	a. solid-liquid b. liquid-vapour c. vapour -solid d. liquid-liquid		
6	-----phase rule is applicable for Ag-Pb alloy system.	2	1
	a. Expanded b. Normal c. Ohms d. Condensed		
7	Calorific value of Peat coal is -----	4	1
	a. 5400 b. 1500 c. 250 d. 8000		
8	Pitting corrosion is ----- type of corrosion.	3	1
	a. localised b. delocalised c. wet d. Dry		
9	-----corrosion occurs due to the action of microbs.	3	1
	a. pitting b. Microbial c. Galvanic d. wet		
10	Which of the following is an example of Solid Lubricant?	4	1
	a Talc b. Petrol c. Diesel d. None		

11	Phenolphthalein is a weak ----- acid.				5	1
	a. Inorganic	b. Organic	c. neutral	d. chemical		
12	Glass electrode is used to measure -----				5	1
	a. resistance	b. conductance	c. electricity	d. pH		
Q. 2	Solve the following.					12
A)	What is Dissolve Oxygen ? How it can be determined by Winkler's Method				1	6
B)	Discuss in brief disadvantages of hard water in Domestic and Industrial use.				1	6
Q.3	Solve the following.					12
A)	Explain the terms Phase and Component with suitable examples in Phase Rule equation.				2	6
B)	Explain in detail one component phase diagram of Water system				2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain methods to minimize the rate of corrosion.				3	6
B)	Discuss Direct or Dry chemical corrosion.				3	6
C)	Write a note on : Sacrificial anodic protection method.				3	6
Q.5	Solve Any Two of the following.				4	12
A)	Discuss different types of Coal with its characteristics and uses.				4	6
B)	Explain in detail Physical and chemical properties of Lubricants.				4	6
C)	Write a note on proximate analysis of coal.				4	6
Q. 6	Solve Any Two of the following.					12
A)	Explain the principal of Conductometric titration with example of a) Strong acid and Strong base b) Weak acid and Strong base.				5	6
B)	Discuss the determination of conductance by Wheatstone's bridge.				5	6
C)	Write a note on Glass electrode.				5	6
*** End ***						